

World-Model-Grounded LLM Planning for AUV and ASV Navigation Near Offshore Wind Farms



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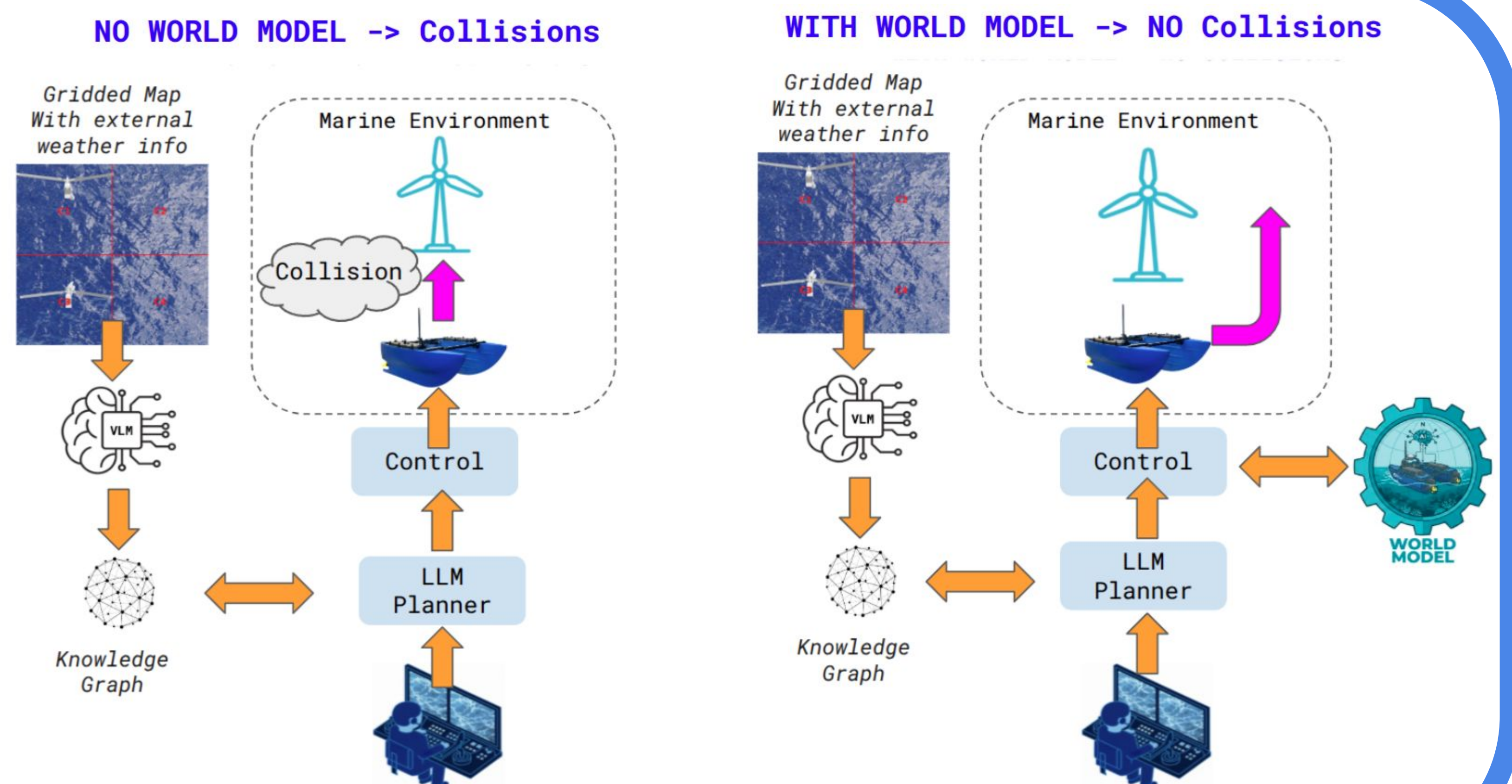
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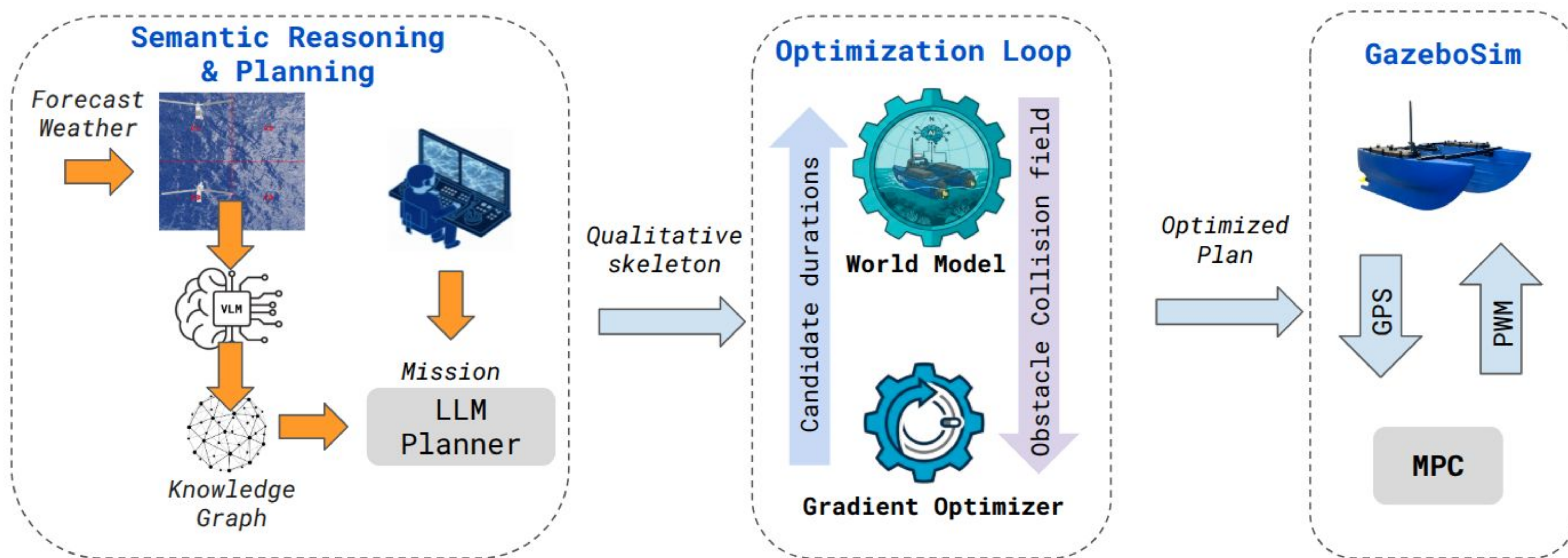
MOTIVATION

Large language models (LLMs) can translate natural-language missions into actions but lack physics awareness, often overlooking drag, thruster dynamics, and currents.

Our framework combines LLMs with a physics-based world model: the LLM decides what to do, while the world model determines safe execution timing.



THE MAIN CONTRIBUTION



1. Physics-Grounded World Model:

Pairs analytical marine dynamics with a learned residual network for both AUV(6-DOF) and ASV(3-DOF) platforms.

2. Gradient Trajectory Optimizer:

Optimizes action step durations through a differentiable collision field without re-querying the LLM.

3. MPC (Model Predictive Controller) with Trust-Region Guard:

Closed-loop replanner feeds back state data while preventing plan collapse from drift.

4. VLM Semantic Mapping:

Extracts obstacle maps directly from satellite images, nautical charts, and weather forecasts.

KEY RESULTS

100% Collision-Free Navigation:

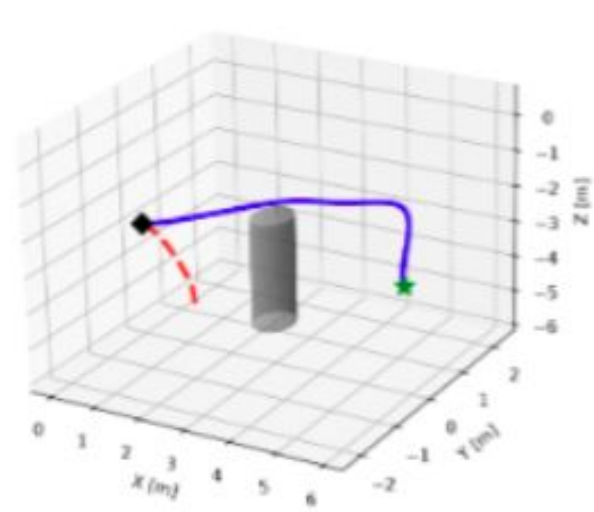
Reached all goals in GazeboSim under waves and currents.

Up to 93% Less Error: Reduced goal-distance error by 70%–82% (ASV) and ~93% (AUV) versus baseline LLMs.

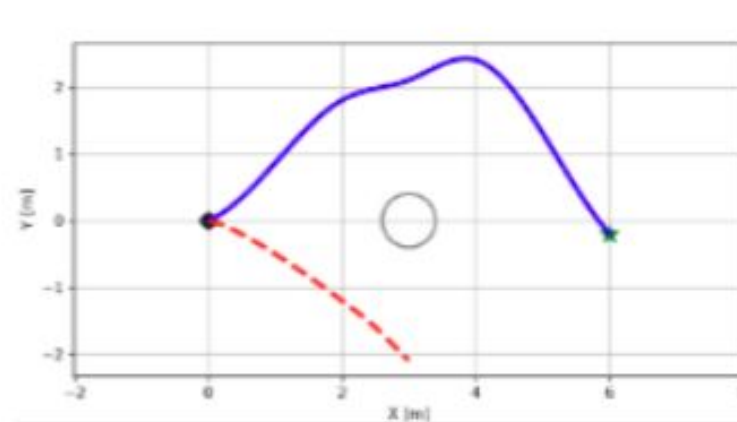
96% Map Accuracy: Matched hand-drawn obstacle maps using only open satellite and marine data.



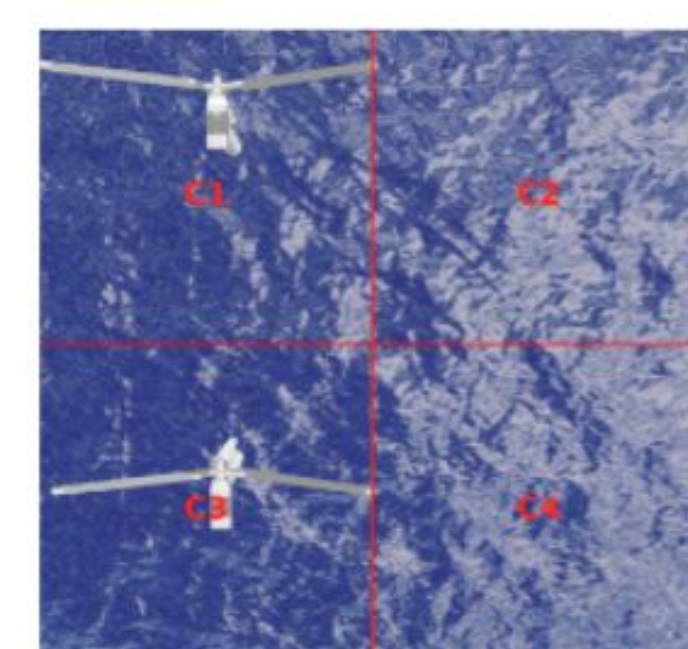
GazeboSim (AUV)



AUV 3-D



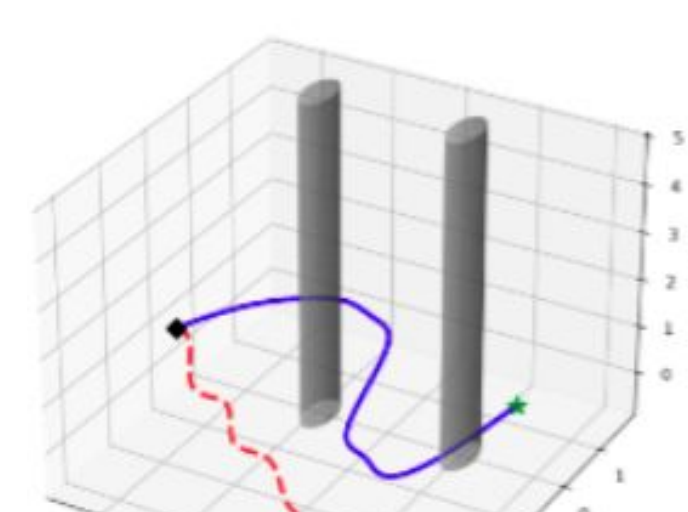
AUV top-down



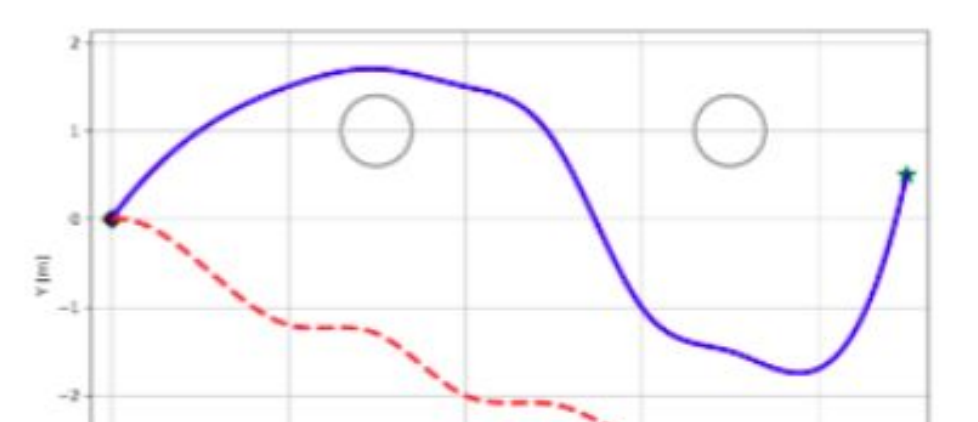
VLM grid cells



GazeboSim (ASV)



ASV 3-D



ASV top-down